

By: AKH  
 Rev.--- By: ---  
 Chk: ---

Date May 2026  
 Date ---  
 Date ---

## Macaulay Test

## Macaulay Beam Analysis - BetaTesting

## 05 - Steel - Triangular Distributed Load

Bridge, L = in

Beam Sect = W14x61

A = 17.90 in<sup>2</sup>I = 640.00 in<sup>4</sup>E = 29,000,000 lb / in<sup>2</sup>G = 1.740E+06 lb / in<sup>2</sup> = 0.06\*29,000,000

Length Units in

Force Units lb

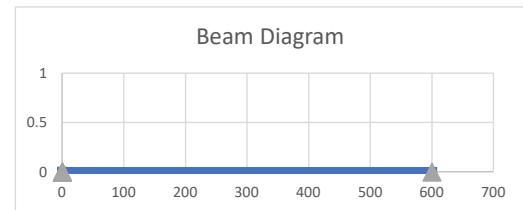
## Beam Segments (First Segment Begins at 0)

|                                    | Segment No. | X End  | A               | I               | E                    | G                    |
|------------------------------------|-------------|--------|-----------------|-----------------|----------------------|----------------------|
|                                    |             | in     | in <sup>2</sup> | in <sup>4</sup> | lb / in <sup>2</sup> | lb / in <sup>2</sup> |
| DO NOT<br>DELETE ROWS<br>(GRAPHED) | 1           | 600.00 | 17.90           | 640.00          | 2.900E+07            | 3.115E+07            |
|                                    | 2           |        |                 |                 |                      |                      |
|                                    | 3           |        |                 |                 |                      |                      |
|                                    | 4           |        |                 |                 |                      |                      |
|                                    | 5           |        |                 |                 |                      |                      |

## Supports

## Prescribed Disp. (typ 0)

|                                    | Supp. No. | X     | dY (vert) | dr (rot) |
|------------------------------------|-----------|-------|-----------|----------|
|                                    |           | in    | in        | rad      |
| DO NOT<br>DELETE ROWS<br>(GRAPHED) | 1         | 0.0   | 0.0       | 0.000    |
|                                    | 2         | 600.0 | 0.0       | 0.000    |
|                                    | 3         |       |           |          |
|                                    | 4         |       |           |          |
|                                    | 5         |       |           |          |



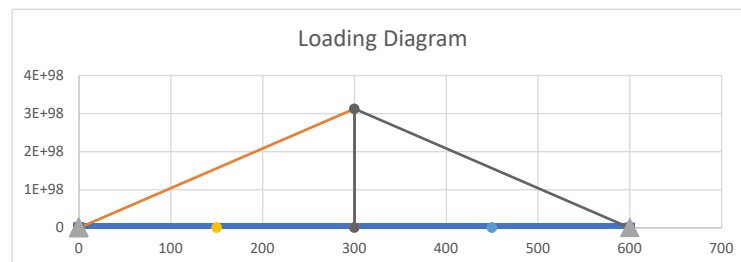
## Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

|                                    | Load No. | x1    | x2    | w1 (at x1) | w2 (at x2) |
|------------------------------------|----------|-------|-------|------------|------------|
|                                    |          | in    | in    | lb / in    | lb / in    |
| DO NOT<br>DELETE ROWS<br>(GRAPHED) | 1        | 0.0   | 300.0 | 0.0        | -6.3       |
|                                    | 2        | 300.0 | 600.0 | -6.3       | 0.0        |
|                                    | 3        | 0.0   | 0.0   | 0.0        | 0.0        |
|                                    | 4        | 0.0   | 0.0   | 0.0        | 0.0        |
|                                    | 5        |       |       |            |            |

| Load    |
|---------|
| lb / ft |
| 75.0    |
| 75.0    |
| 0.0     |
| 0.0     |

## Point Loads - UNFACTORED

|                                    | Load No. | X   | Py   | My    |
|------------------------------------|----------|-----|------|-------|
|                                    |          | in  | lb   | lb-in |
| DO NOT<br>DELETE ROWS<br>(GRAPHED) | 1        | 150 | 0.00 | 0.00  |
|                                    | 2        | 450 | 0.00 | 0.00  |
|                                    | 3        |     |      |       |
|                                    | 4        |     |      |       |
|                                    | 5        |     |      |       |



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#### BEAM REACTIONS

| Position (X) | Enforced Disp (dy) | Enforced Rot (theta) | Vert Reac (Ry) | Internal Moment (Mz) | Note             |
|--------------|--------------------|----------------------|----------------|----------------------|------------------|
| (L)          | (L)                | (rad)                | (F)            | (F-L)                | (---)            |
| 0.00         | 0.0                | 0.0                  | 937.5          | 0.0                  | Support reaction |
| 600.00       | 0.00               | 0.00                 | 937.5          | 0.0                  | Support reaction |

| Vert Reac |
|-----------|
| (kip)     |
| 0.938     |
| 0.938     |

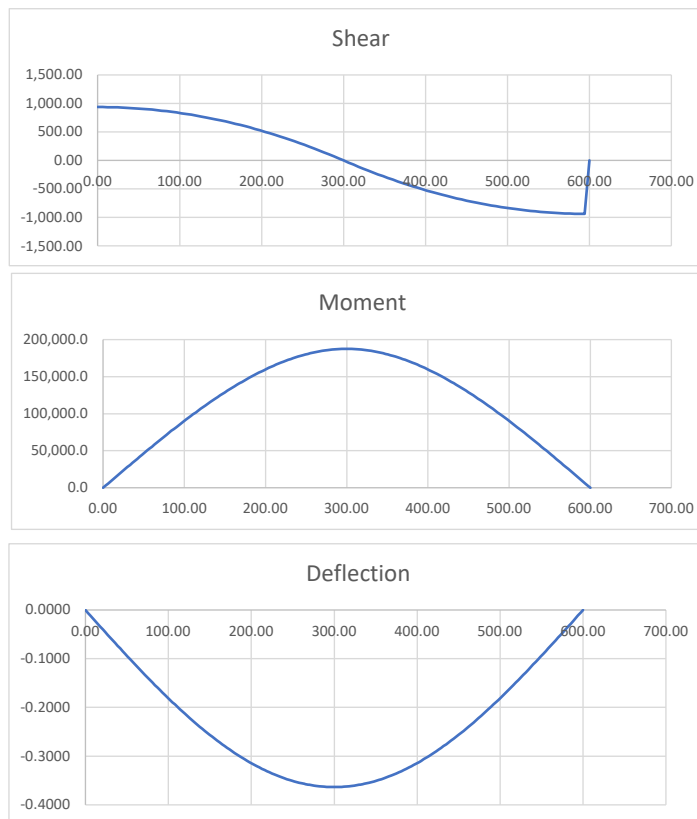
#### BEAM REACTIONS EQUILIBRIUM CHECK

| Item        | Units | Loads       | Reactions  | Residual | Check Sum |
|-------------|-------|-------------|------------|----------|-----------|
| Sum Fy      | F     | -1,875.0    | 1,875.0    | 0.0      | OK        |
| Sum M @ x=0 | F-L   | -5.6250E+05 | 5.6250E+05 | 0.0      | OK        |

#### BEAM MAX/MIN LOAD EFFECTS

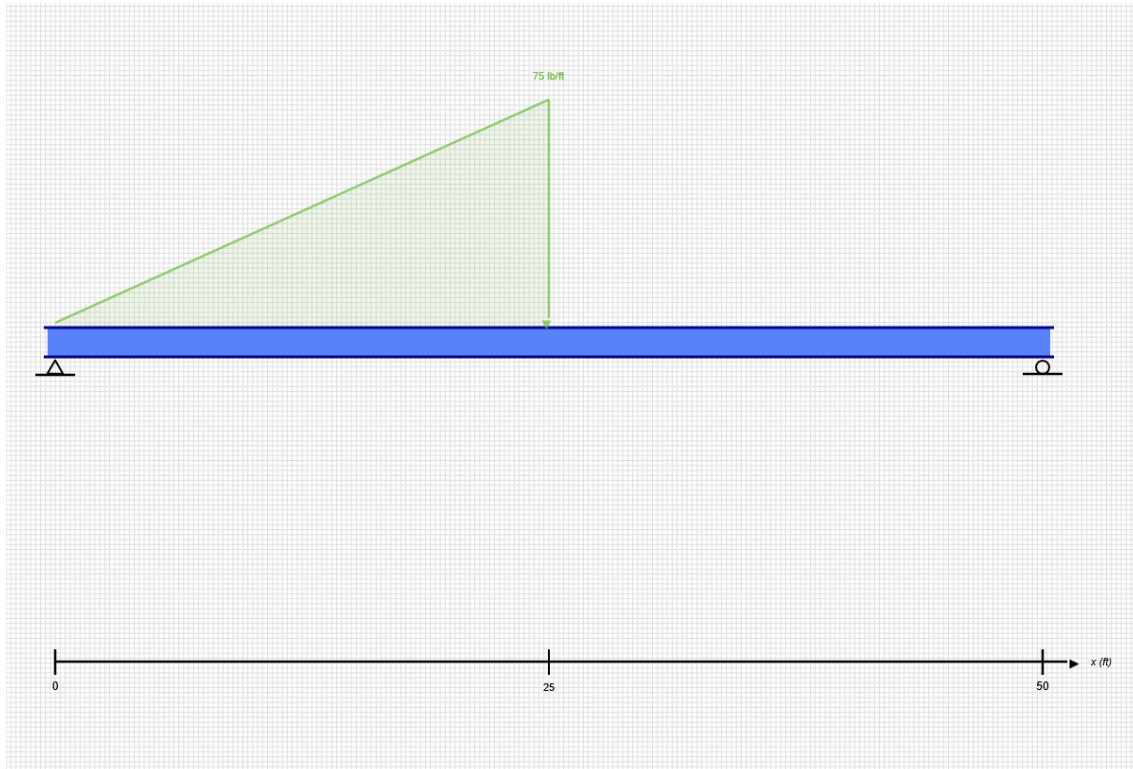
| Item     | Position (X) | Value      |
|----------|--------------|------------|
| (---)    | (L)          | (F or F-L) |
| Max +V   | 0.00         | 937.5      |
| Max -V   | 594.00       | -937.1     |
| Max +M   | 300.00       | 187,500.0  |
| Max -M   | 0.00         | 0.0        |
| Max Defl | 300.00       | -0.364     |

| Value        |
|--------------|
| (---)        |
| 0.94 kip     |
| -0.94 kip    |
| 15.63 kip-ft |
| 0.00 kip-ft  |
| -0.364 in    |



## SKYCIV BEAM ANALYSIS REPORT

Load Combination: DL



Software: SkyCiv Beam v3.3.2  
Tue Jun 09 2026 14:08:40 GMT-0400 (Eastern Daylight Time)

### Project Info

File Name: 01-Steel-Beam

### Included in this Report:

- Input Summary
- Beam Section
- Free Body Diagram (FBD)
- Analysis Summary
- Analysis Results
- Bending Moment Diagram (BMD)
- Shear Force Diagram (SFD)
- Deflection Results

## INPUT SUMMARY

### General Info

|               |        |
|---------------|--------|
| Beam Length:  | 50 ft  |
| Section Name: | W14x61 |
| Self Weight:  | False  |

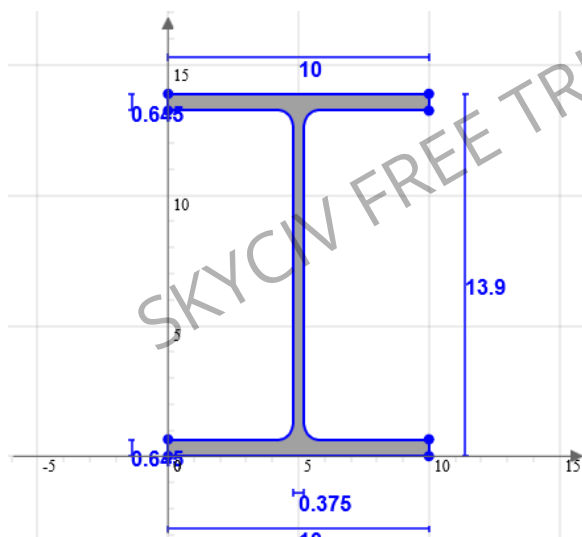
### Supports

| Support Type | Location |
|--------------|----------|
| Pinned       | 0 ft     |
| Roller       | 50 ft    |

### Loads

| Load Type        | Location      | Magnitude      | Load Case |
|------------------|---------------|----------------|-----------|
| Distributed Load | 0 ft to 25 ft | 0 lb to -75 lb | DL        |

### Beam Section



| Geometric Properties |      |                 |
|----------------------|------|-----------------|
| A                    | 17.9 | in <sup>2</sup> |
| C <sub>z</sub>       | 5    | in              |
| C <sub>y</sub>       | 6.95 | in              |

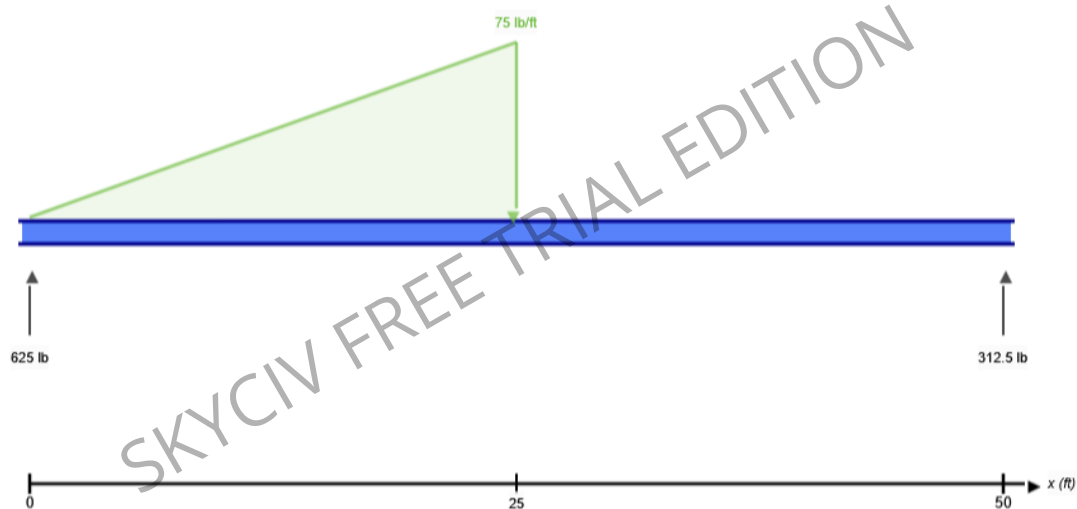
| Bending Properties |     |                 |
|--------------------|-----|-----------------|
| I <sub>z</sub>     | 640 | in <sup>4</sup> |
| I <sub>y</sub>     | 107 | in <sup>4</sup> |

| Shear Properties |        |                 |
|------------------|--------|-----------------|
| A <sub>z</sub>   | 11.277 | in <sup>2</sup> |
| A <sub>y</sub>   | 4.99   | in <sup>2</sup> |

| Torsion Properties |       |                 |
|--------------------|-------|-----------------|
| J                  | 2.19  | in <sup>4</sup> |
| r                  | 0.991 | in              |

| Shape  | Material         | E (ksi) | v    | ρ (lb/ft <sup>3</sup> ) |
|--------|------------------|---------|------|-------------------------|
| W14x61 | Structural Steel | 29000   | 0.27 | 490                     |

### FREE BODY DIAGRAM



### RESULT SUMMARY

| Check               | Status | Limit  | Ratio | Max       |
|---------------------|--------|--------|-------|-----------|
| Deflection          | PASS   | L/250  | 0.076 | L/3284    |
| Custom Stress Limit | PASS   | 39 ksi | 0.028 | 1.108 ksi |
| Material Yield      | PASS   | 36 ksi | 0.031 | 1.108 ksi |
| Material Strength   | PASS   | 58 ksi | 0.019 | 1.108 ksi |

## ANALYSIS RESULTS

### Reactions

| Support at | R <sub>x</sub> (lb) | R <sub>y</sub> (lb) | M (kip-ft) |
|------------|---------------------|---------------------|------------|
| 0          | 0                   | 625                 | 0          |
| 50         | 0                   | 312.5               | 0          |

### Force Extremes

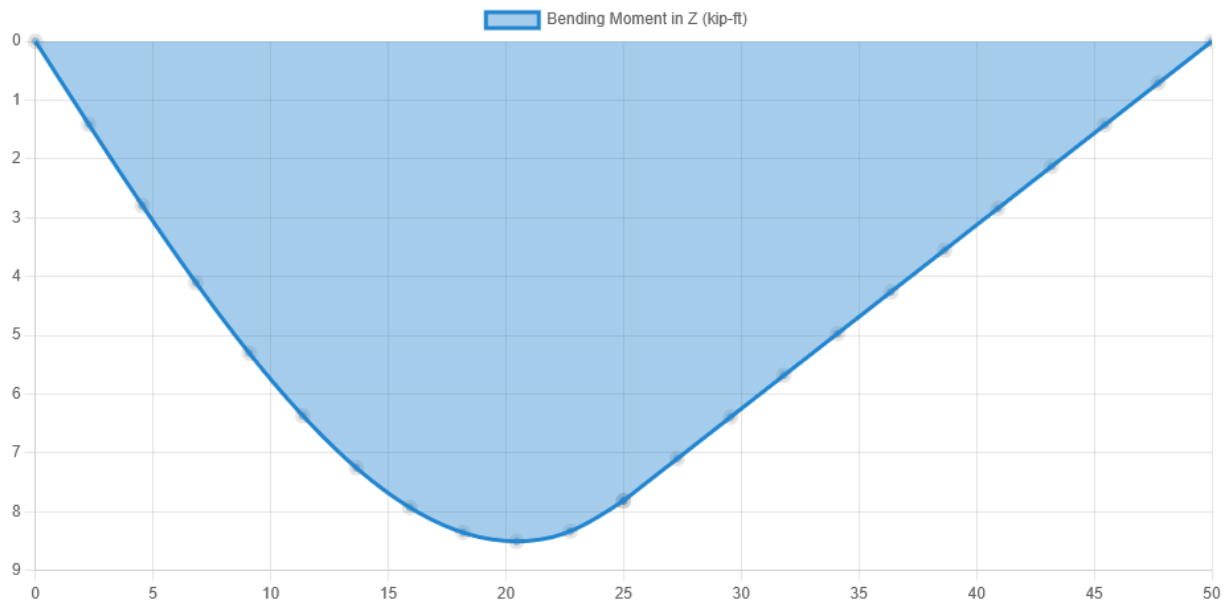
| Result         | Max          | Min       |
|----------------|--------------|-----------|
| Bending Moment | 8.505 kip-ft | 0 kip-ft  |
| Shear          | 625 lb       | -312.5 lb |
| Displacement Y | 0 in         | -0.183 in |

### Stress Extremes

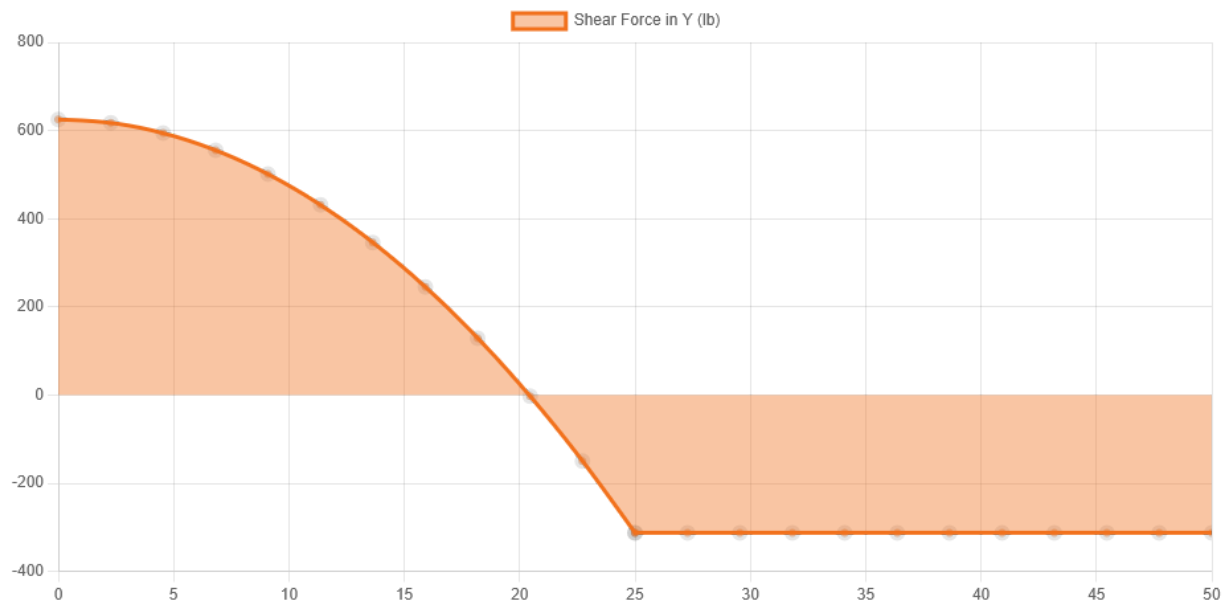
| Result                     | Max (ksi) | Min (ksi) |
|----------------------------|-----------|-----------|
| Bending Stress             | 1.108     | -1.108    |
| Shear Stress Total         | 0.133     | 0.001     |
| Max Combined Normal Stress | 1.108     | 0         |
| Min Combined Normal Stress | 0         | -1.108    |

## DIAGRAMS

### Bending Moment Diagram



### Shear Force Diagram



Displacement



|                                                                                                                                      |     |         |     |
|--------------------------------------------------------------------------------------------------------------------------------------|-----|---------|-----|
| Location (ft)                                                                                                                        | 0   | 22.727  | 50  |
| Total Deflection (in)                                                                                                                | 0 ? | 0.183 ? | 0 ? |
| Deflection/Span                                                                                                                      | -   | L/3284  | -   |
| ⓘ The Deflection/Span results are calculated using the analysis results and the Deflection Limit of L/250 set in the model settings. |     |         |     |